Prob. 8. (2 & T, Ch. 7, 2.7 a, b)

(a) Use separation of vars to show that the following are (the only) <u>bounded</u> <u>harmonic</u> f-ns in the upper half plane, $y>0$ in $\mathbb{R}^2$, (obtained by separation of variables):

(1) $$e^{-\lambda y}\cos(\lambda x), \quad e^{-\lambda y}\sin(\lambda x); \qquad \lambda \geq 0.$$

(b) By using superposition principle, show that by integrating with respect to (wrt) parameter λ, we obtain two more f-ns:

(2) $$\frac{x}{x^2+y^2}, \quad \frac{y}{x^2+y^2}$$

which are also bounded harmonic functions in the upper half plane, $y>0$ in $\mathbb{R}^2$.

Sol-n.

(a) Note that, by Prop. ① of harmonic f-n $u = u(x,y)$,

Thm 7.1.

Let u be a cont-y sol-n of Laplace's eq-n in a domain Ω. Then u is analytic in Ω.

Def-n. (Harmonic f-n)
u is Harmonic if $u \in C^2(\Omega)$ & $\Delta u = 0$.

(1) Let $u(x,y) = X(x) \cdot Y(y)$.

Then $u_{xx} = X''(x) \cdot Y(y)$ &
$u_{yy} = X(x) \cdot Y''(y)$.

[For some $X(x) \neq 0$ & $Y(y) \neq 0$.
If $X(x) = 0$ or $Y(y) = 0$
then $u(x,y) = 0 \Rightarrow \Delta u = 0$
trivially. ☆]

Then

$$\Delta u = u_{xx} + u_{yy} = X''(x) \cdot Y(y) + X(x) \cdot Y''(y) = 0 \Rightarrow$$

(2) $$X''(x) \cdot Y(y) = -X(x) \cdot Y''(y)$$ [By ☆, $X(x) \neq 0$, $Y(y) \neq 0$]

(3) $$\underbrace{\frac{X''(x)}{X(x)}}_{\text{f-n of only } x.} = \underbrace{-\frac{Y''(y)}{Y(y)}}_{\text{f-n of only } y} = \mu = \text{const.} \in \mathbb{R}$$

(4) $$\frac{X''(x)}{X(x)} = \mu$$

(5) $$-\frac{Y''(y)}{Y(y)} = \mu$$

Now we should solve (4) + (5). Let's see first

1) $\mu > 0$

$$\frac{X''}{X} = \mu > 0$$

(6) $$X'' - \mu X = 0.$$

Assume the L.I sol-n for (6) is in the form of

$X = e^{\lambda x}$, then $X' = \lambda e^{\lambda x}$ & $X'' = \lambda^2 e^{\lambda x}$ $\overset{(6)}{\Rightarrow}$

$$\lambda^2 e^{\lambda x} - \mu \cdot e^{\lambda x} = 0 \quad \Big\} : e^{\lambda x} \neq 0$$

$$\lambda^2 - \mu = 0 \qquad (\mu > 0 \text{ recall, })$$

(7) $$\boxed{\lambda = \pm\sqrt{\mu}}$$

Then, Fundamental solution set (F.S.S) is

(8) $$F.S.S(X) = \left\{ e^{\sqrt{\mu} x}, \; e^{-\sqrt{\mu} x} \right\}$$

Then, the $Y(y)$ eq-n in (5)

$$\frac{Y''(y)}{Y(y)} = -\mu \qquad (\mu > 0)$$

(9) $$Y'' + \mu Y = 0$$

Assuming $Y = e^{\lambda y}$, $\Rightarrow Y' = \lambda e^{\lambda y}$, $Y'' = \lambda^2 e^{\lambda y}$, (9) is

$$\lambda^2 e^{\lambda y} + \mu \cdot e^{\lambda y} = 0 \quad \Big\} : e^{\lambda y} \neq 0$$

$$\lambda^2 + \mu = 0 \qquad \lambda = \pm i\sqrt{\mu}$$

then the F.S.S for (9) is

(10) $$F.S.S(Y) = \{ \cos(\sqrt{\mu}\, y),\ \sin(\sqrt{\mu}\, y) \}.$$

Then by (8) & (10), F.S.S for $u(x,y) = X(x) \cdot Y(y)$ is

(11) $$F.S.S(u) = \{ e^{\sqrt{\mu}x} \cos(\sqrt{\mu}\, y),\ e^{\sqrt{\mu}x} \sin(\sqrt{\mu}\, y),\ e^{-\sqrt{\mu}x} \cos(\sqrt{\mu}\, y),\ e^{-\sqrt{\mu}x} \sin(\sqrt{\mu}\, y) \}$$

Then, second,

2) $\mu < 0$. Then let $\tilde{\mu} = -\mu$, then (4) & (5) become

$$\frac{X''}{X} = -\tilde{\mu} \quad (\tilde{\mu} > 0) \quad \& \quad \frac{Y''}{Y} = \tilde{\mu} \quad (\tilde{\mu} > 0)$$

$X'' + \tilde{\mu} X = 0$ $\qquad$ $Y'' - \tilde{\mu} Y = 0$

$\Downarrow$ $X = e^{\lambda x}$ $\qquad$ Assume $Y = e^{\lambda x}$

$\lambda^2 + \tilde{\mu} = 0$ $\qquad$ $\lambda^2 - \tilde{\mu} = 0$, $\lambda = \pm\sqrt{\tilde{\mu}}$

$$F.S.S.(X) = \{ \cos(\sqrt{\tilde{\mu}}\, x),\ \sin(\sqrt{\tilde{\mu}}\, x) \};\quad F.S.S.(Y) = \{ e^{\sqrt{\tilde{\mu}}\, y},\ e^{-\sqrt{\tilde{\mu}}\, y} \}$$

Then

(12) $$F.S.S.(u) = \{ e^{\sqrt{\tilde{\mu}}\, y} \cos(\sqrt{\tilde{\mu}}\, x),\ e^{\sqrt{\tilde{\mu}}\, y} \sin(\sqrt{\tilde{\mu}}\, x),\ e^{-\sqrt{\tilde{\mu}}\, y} \cos(\sqrt{\tilde{\mu}}\, x),\ e^{-\sqrt{\tilde{\mu}}\, y} \sin(\sqrt{\tilde{\mu}}\, x) \}$$

And finally,

3) $\mu = 0$.

$$\frac{X''}{X} = 0 \ (\mu = 0) \qquad \frac{Y''}{Y} = 0 \ (\mu = 0)$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$X'' = 0 \qquad\qquad Y'' = 0$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$X(x) = C_1 x + C_2 \qquad\qquad Y(y) = D_1 y + D_2$$

$$F.S.S.(X) = \{1, x\} \qquad\qquad F.S.S.(Y) = \{1, y\}$$

Then, $u = X(x) \cdot Y(y)$

(13) $F.S.S.(u) = \{1, y, x, x \cdot y\}$, but none of the solutions in (13) is bounded in the upper half plane, $y > 0$ in $\mathbb{R}^2$, except for $\boxed{u(x,y) = 1.}$ (14)

By (12), the bounded sol-ns in the upper half plane, $y > 0$, are

(15) $\boxed{e^{-\sqrt{\tilde{\mu}}\, y} \cdot \cos\left(\sqrt{\tilde{\mu}}\, x\right)}$, $\boxed{e^{-\sqrt{\tilde{\mu}}\, y} \cdot \sin\left(\sqrt{\tilde{\mu}}\, x\right)}$ (16) for $\tilde{\mu} > 0$.

By (11), there are NO bounded sol-ns because of $e^{\pm\sqrt{\mu} \cdot x}$ where $x \in \mathbb{R}$, not bounded.

But (14), (15), (16) can be combined into the following, by denoting

$$\sqrt{\tilde{\mu}} = \lambda \quad , \quad \lambda \geq 0 \quad :$$

(14) $\quad F.S.S(u) = \left\{ e^{-\lambda y} \cos(\lambda x), \quad e^{-\lambda y} \cdot \sin(\lambda x) \right\}, \quad \lambda \geq 0$

Note if we let $\lambda = 0$, we get $u(x,y) = 1$ in (14). The F.S.S(u) in (14) is bounded in the upper half plane & satisfy the harmonicity prop s. $\blacksquare$